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Regulation of macrophage Fc receptor expression and phagocytosis by histidine-rich glycoprotein

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SUMMARY

Regulation of macrophage Fc receptor (FcγR)-mediated phagocytic function by histidine-rich glycoprotein (HRG) was investigated. Pretreatment of oil-elicited inflammatory mouse peritoneal macrophages with HRG for 1–3 hr increased their FcγR-mediated binding and phagocytosis of IgG-opsonized sheep erythrocyte conjugates (EA). A significant reduction of FcγR-dependent EA binding and phagocytosis occurred after pretreatment of macrophages with HRG for more than 8 hr. These results indicate that HRG is capable of modulating FcγR expression in a biphasic fashion, which directly affects the overall efficiency of phagocytosis. HRG differentially regulated the functions of FcγR subclasses. For example, HRG reduced the efficiency of FcγRII (Fcγ2b/γ1R)-dependent phagocytosis of erythrocytes conjugated with monoclonal IgG2b or IgG1 by macrophages pretreated with HRG for 24 hr. However, when similar studies were performed using erythrocytes coated with monoclonal IgG2a, HRG was less effective in inhibiting FcγRI (Fcγ2aR)-dependent phagocytosis. As an HRG-binding glycosaminoglycan, heparin failed to block the regulatory function of HRG on macrophages. Similarly, interferon-γ (IFN-γ) was not capable of blocking the functional activity of HRG. These studies suggest that HRG regulates macrophage function via a novel pathway different from that of heparin or IFN-γ.

INTRODUCTION

Histidine-rich glycoprotein (HRG) is a plasma glycoprotein that consists of unusually high contents of histidine and proline.[1,2] Although a number of biological properties of HRG has been reported, its physiological function has not been well established. HRG is known to bind divalent cations[3] and heparin,[4,5] and interact with plasma thrombospondin[6] and plasminogen.[2] HRG is capable of inhibiting contact activation of blood coagulation.[7] Recently it was reported that HRG complexes with serum S-protein/vitronectin and modulates complement functional efficiency of both classical and alternative pathways.[8]

HRG is involved in regulation of immune cellular functions, in that it inhibits autorosette formation between T lymphocytes and autologous erythrocytes.[9–11] Saigo *et al.*[12] determined that HRG binds to a 56,000 MW receptor on T cells. The binding appears to provide a signal for suppression of T-cell proliferation in response to stimulation by interleukin-2 (IL-2) and antibodies against antigen receptor (CD3).[13] In this study, it is shown that HRG is capable of biphasic modulation of macrophage Fcγ receptor (FcγR) expression and phagocytosis. Mechanisms by which HRG regulates FcγR expression and function are discussed.

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MATERIALS AND METHODS

Purification of plasma HRG

The purification strategy was initially designed to isolate novel plasma protein(s) that would modulate macrophage Fc expression and phagocytic function. Plasma mannose-type glycoproteins which are specific for binding to the lectin, concanavalin A (Con A), were found to possess the activity of regulating macrophage phagocytic functions. A further chromatographic step utilizing a heparin column revealed that a major glycoprotein which bound strongly to heparin modulated macrophage function. This glycoprotein was further purified by ionic-exchange chromatography. Subsequent gas-phase microsequencing for determining N-terminal sequence revealed that this glycoprotein was identical to plasma HRG. Overall, the purification of plasma HRG was performed as follows: mouse blood was collected by cardiac puncture from male 8–12-week-old C3HeB/FeJ mice (Jackson Laboratory, Bar Harbor, ME), and mixed with an equal volume of 5 mM barbital-buffered saline containing 10 mM EDTA.[14] Firstly, affinity chromatography was performed to isolate plasma mannose-type glycoproteins. One hundred millilitre EDTA-treated mouse plasma was applied to a 2·5 × 13 cm Con A–Sepharose 4B column (Sigma Chemical Co., St Louis, MO; 50 ml bed volume), which was pre-equilibrated with a 50 mM Tris-HCl buffer, pH 8·0, containing 100 mM NaCl, followed by washing with 10-bed volumes of the equilibration buffer. The column-bound proteins were eluted using 2% α-D-mannopyranoside (Sigma), concentrated by ultrafiltration using an Amicon PM 10 membrane (MW exclusion of 10,000; Amicon Corporation, Danvers, MA) and then dialysed against 0·03 M sodium phosphate buffer, pH 7·0, containing 0·2% sodium azide. Secondly, the pooled fractions were chromatographed on a 2·5 × 13 cm heparin–agarose affinity column (Sigma; 50 ml bed volume) equilibrated with the 0·03 M sodium phosphate buffer. The HRG protein fractions were eluted from the column using a linear salt gradient of 0·5–1 M NaCl. The pooled fractions were dialysed against the 0·03 M phosphate buffer and finally chromatographed on a DEAE–Sephacel column (2·5 × 15 cm; Sigma). By applying a linear gradient of 0·0–0·5 M NaCl to the column, the purified HRG was obtained at the conductivity ranges of approximately 5·0–10·0 mmho. Protein purity was analysed in both 7 and 10% SDS–PAGE, and quantified by the BioRad protein assay (BioRad Laboratories, Richmond, CA). Contamination with bacterial lipopolysaccharides was assessed using a chromogenic limulus amoebocyte lysate microassay of Whittaker, M.A. Bioproducts (Walkersville, MO). The purified HRG preparations were exhaustively dialysed against RPMI-1640 medium prior to applying for experiments. Isolation of human HRG from EDTA-treated or citrated normal human plasmas was performed identically to the above procedures.

Macrophage FcγR-mediated phagocytosis

Photometric measurement of mouse inflammatory peritoneal macrophage FcR-mediated phagocytosis of antibody-conjugated sheep erythrocytes (EA) was performed as previously described.[15–18] Briefly, male C3HeB/FeJ mice were injected with 2 ml sterile paraffin oil (Fisher Scientific Co., Pittsburgh, PA) 3 days prior to harvest of the inflammatory peritoneal macrophages. Peritoneal non-inflammatory resident macrophages were also harvested from control mice. Macrophages (2×10^6/ml), suspended in RPMI-1640 supplemented with 200 mM L-glutamine, 25 mM HEPES, and 50 μg/ml gentamycin (Gibco, Grand Island, NY), were seeded in 100-μl volumes into individual wells of 96-well flat-bottom microtitre plates (Corning Glass Works, Corning, NY). Following adherence for 1 hr at 37°, the non-adherent cells were removed by washing with RPMI-1640 medium. EA was prepared by conjugation of various subagglutinating dilutions of rabbit anti-sheep erythrocyte polyclonal IgG (Cordis Laboratories, Miami, FL) with an equal volume of 5% erythrocytes (Colorado Serum Co., Denver, CO) at 37° for 30 min. EA cells were washed twice and further diluted to 0·5% prior to use. Sheep erythrocytes (E) without antibody conjugation were used as controls. Macrophages were pretreated with or without HRG at 4° or 37° for various indicated times. HRG was then decanted from the microtitre plates, and the cell layers were washed twice with RPMI-1640 medium prior to adding EA cells. Alternatively, HRG was allowed to remain in the medium for continuous co-incubation with EA cells. Binding and/or phagocytosis of E or EA cells (100 μl of the 0·5% cell stocks) by macrophages was carried out at 37° for 1 hr. Unless otherwise specified, all the experiments were performed under serum-free conditions to prevent possible interactions between HRG and serum proteins or other factors.[2–8] After 1 hr incubation, the macrophage monolayers were washed with RPMI-1640 to remove unbound EA, followed by methanol fixation, air drying and quantification of the total index, which included EA binding and internalization (phagocytosis) by macrophages using a MR580 reader (Dynatech, Alexandria, VA) at 405 nm for the absorbance of haemoglobin. For determination of the phagocytic index, the washed monolayers were treated with hypotonic saline (0·09%) to lyse extracellular bound EA prior to methanol fixation, air drying and quantification of internalized EA. Where indicated, FcγR-dependent binding of EA was also carried out by exposure of macrophages to EA for 1–2 hr at 4°. Both phagocytic and total indices were calculated as:

$$1 - [E_{405}/EA\ (IgG)_{405}] \times 100 = \text{Index}.$$

Murine monoclonal antibodies, IgG2a and IgG2b against sheep erythrocytes produced from hybridoma cell lines SS.1 and NS8.1, respectively [American Type Culture Collection (ATCC), Rockville, MD], were also used for preparing opsonized erythrocyte conjugates for phagocytic studies. Monoclonal IgG1 against sheep erythrocytes was obtained from Accurate Chemical (Westbury, NY.) To determine the viability of macrophages during 1–48 hr under serum-free culture conditions, ^{51}Cr-release assays were performed with control and HRG-treated cells.[19]

RESULTS

HRG isolation and characterization

Figure 1 depicts the sequential chromatographic profiles for the isolation of HRG from human citrated plasma. Similar chromatographic patterns were obtained when using mouse plasma (data not shown). Approximately 3·0% of the total plasma proteins applied to the Con A–Sepharose 4B column was recovered initially (Fig. 1a). The second heparin–agarose (Fig. 1b) and the final DEAE–Sephacel (Fig. 1c) steps yielded 0·015 and 0·0045%, respectively, of the total plasma proteins applied. Purified HRG was obtained from the DEAE–Sephacel column at the conductivity ranges of approximately 5·0–10·0 mmho (Fig. 1c, fraction 1). However, there still remained a minor quantity of HRG along with several major contaminants bound to the DEAE–Sephacel column (Fig. 1c, fractions 2 and 3).

The overall purification scheme was similar to those reported procedures for HRG purification, which involved DEAE ionic-exchange and heparin-affinity chromatography.[3,20] The purified human HRG was subjected to amino-terminal gas-phase microsequencing and the determined partial N-terminal sequence (20 amino acid residues) was found to be identical to the sequence derived from cDNA.[1] In Western blot analysis, rabbit antisera against human serum HRG cross-reacted with mouse-purified serum HRG (data not shown). The purified HRG was not contaminated with bacterial lipopolysaccharides (< 0·01 ng/ml) as confirmed by the limulus amoebocyte assay.

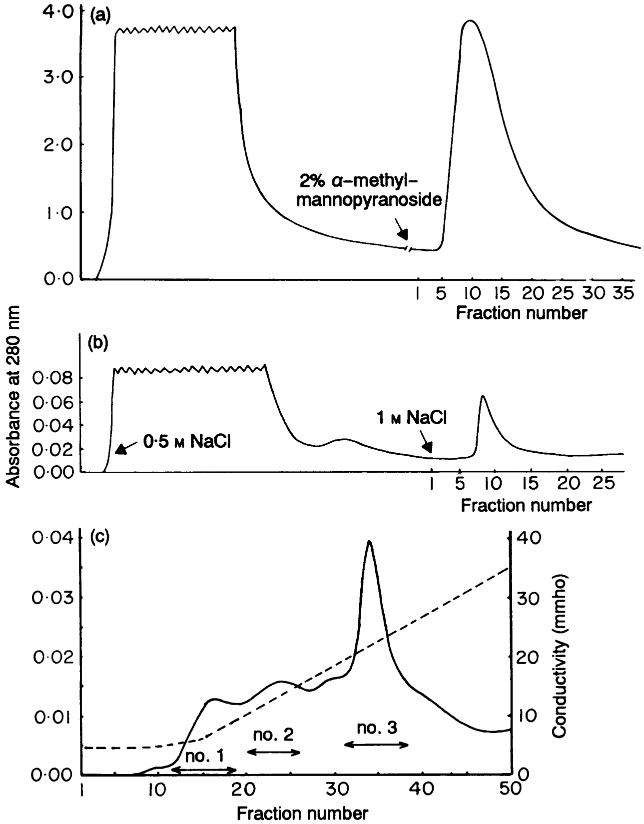

Figure 1. Isolation of HRG from human citrated plasma. One hundred millilitres of plasma was applied to a Con A–Sepharose 4B column and the bound proteins were eluted using 2% α-methyl-mannopyranoside (a, see arrow). The collected fractions were pooled and loaded onto a heparin–agarose column, which was then washed with 0·5 M NaCl before HRG was eluted using 1 M NaCl (b, see arrow). Subsequently, the HRG protein pool was chromatographed on a DEAE-Sephacel column, and a 0·0–0·5 M NaCl linear gradiant was used for eluting bound proteins (c). Purified HRG was obtained at an ionic strength of approximately 5·0–10·0 mmho [see fraction no. 1(c)]. Minor amounts of HRG together with other proteins bound to the column were eluted using relative high salt [see fraction nos. 2 and 3(c)].

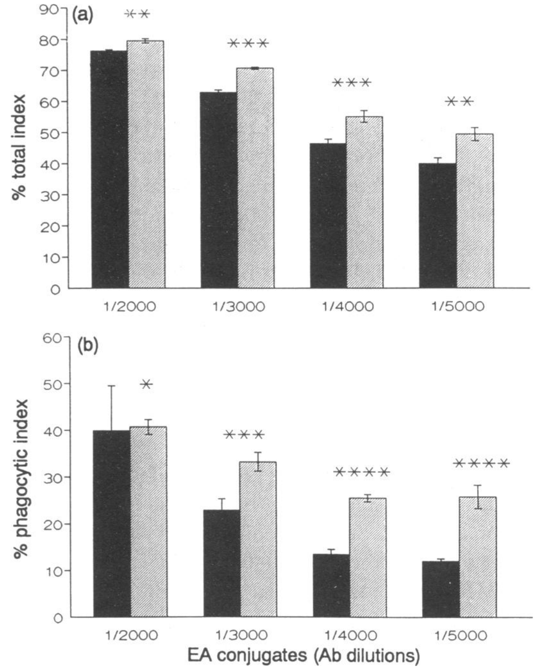

Figure 2. Effect of mouse HRG on macrophage phagocytosis. Oil-elicited mouse peritoneal inflammatory macrophages were pretreated with (▨) or without (■) HRG (125 ng/ml) for 1 hr prior to exposure for 1 hr at 37° to erythrocytes conjugated with various dilutions of polyclonal IgG (EA; IgG 1/2000, 1/3000, 1/4000, and 1/5000 dilutions). Each bar represents the mean ± standard deviation obtained from four experiments (experimental number, $n=4$). Indices were derived from the absolute values (OD at 405 nm) (see Materials and Methods). (a) % total index (binding and phagocytosis); (b) % phagocytic index. Absolute values for the control groups (without HRG treatment) in: (a) EIgG 1/2000, 0·256 ± 0·005; EIgG 1/3000, 0·164 ± 0·004; EIgG 1/4000, 0·114 ± 0·003; EIgG 1/5000, 0·101 ± 0·004; E, 0·061 ± 0·002; (b) EIgG 1/2000, 0·088 ± 0·006; EIgG 1/3000, 0·069 ± 0·002; EIgG 1/4000, 0·061 ± 0·001; EIgG 1/5000, 0·060 ± 0·001; E, 0·053 ± 0·002. *P*-values: experiments versus controls with Student's *t*-tests: *$P > 0·05$; **$0·01 < P < 0·05$; ***$0·001 < P < 0·01$; **** $P < 0·001$.

Regulation of macrophage FcγR-mediated binding and phagocytosis

When the freshly harvested inflammatory peritoneal macrophages were pretreated with mouse HRG (125 ng/ml) for 1 hr at 37° prior to co-incubation with EA, there were increases in FcγR-mediated binding and phagocytosis (Fig. 2). Using Student's *t*-tests, the increases in EA binding and phagocytosis differed significantly from the controls (Fig. 2), except with E that had been conjugated with low saturating dilutions of IgG. Further enhancement of phagocytosis by HRG was observed after 2 hr pretreatment (data not shown). Removal of HRG from the culture after 1–2 hr pretreatment prior to adding EA also resulted in phagocytic enhancement. However, pretreatment of macrophages wth HRG at 4° for 1–2 hr followed by co-incubation with EA for 1 hr at 37° did not result in increases of EA binding and phagocytosis.

The HRG-mediated enhancement of EA binding and phagocytosis was only a transient phenomenon. Reduction of FcγR-mediated phagocytosis was observed after treating macrophages for longer than 8 hr. Figure 3a shows a significant reduction ($P < 0·001$) of macrophage FcγR-mediated phagocytosis of EA conjugates prepared with various antibody dilutions when macrophages were treated with HRG (125 ng/ml) for 20 hr at 37°. In addition, by treating macrophages with 62·5–1000 ng/ml HRG for 24 hr, macrophages lost their ability to phagocytose EA in a dose-related manner (Fig. 3b). Binding of EA conjugates to the 24-hr HRG-treated macrophages at 4° for 1 hr also revealed a dose-dependent reduction of EA binding compared to the untreated control macrophages (Fig. 3b). Similar results were obtained when testing the EA binding at 37°. Pretreatment of macrophages for 1 or 24 hr at 37° with HRG, followed by removing HRG and continued culture in serum-free conditions for an additional 24 hr, also resulted in a reduction of FcγR-mediated phagocytosis.

Although the experimental conditions were performed under serum-free conditions, macrophages did not appear to lose viability during 24–48 hr in culture as determined by ^{51}Cr-release assays (data not shown). Notably, in the presence of 10% foetal bovine serum (FBS), exogenous HRG was also found to reduce macrophage phagocytic functions after 18–24 hr treat-

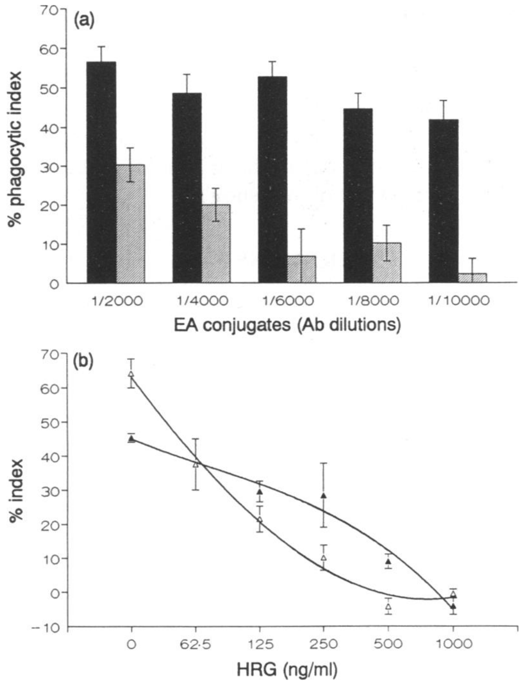

Figure 3. Down-regulation of macrophage phagocytosis by long-term treatment with HRG. (a) Inflammatory macrophages were untreated (■) or pretreated with HRG (■) (125 ng/ml) for 20 hr at 37° prior to the standard phagocytic assay using sheep erythrocytes conjugated with various dilutions of antibodies ($n=4$; mean ± standard deviation). HRG significantly reduced EA phagocytosis by macrophages ($P<0{\cdot}001$; experiments versus controls). Absolute values for the control groups: EIgG 1/2000, 0·172 ± 0·017; EIgG 1/4000, 0·146 ± 0·015; EIgG 1/6000, 0·159 ± 0·014; EIgG 1/8000, 0·136 ± 0·010; EIgG 1/10,000, 0·129 ± 0·012; E, 0·075 ± 0·001. (b) HRG down-regulated macrophage phagocytosis in a dose-related manner (△). Binding of EA cells to control or 24-hr HRG-treated macrophage at 4° for 1 hr also revealed a reduction of EA binding ($n=4$; mean ± standard deviation) (▲). Absolute values for the control groups: (1) phagocytosis, EA, 0·155 ± 0·022; E, 0·056 ± 0·002; (2) 4° binding, EA, 0·116 ± 0·004; E, 0·064 ± 0·002.

ment (data not shown). The freshly harvested peritoneal macrophages were still responsive to the HRG-mediated biphasic modulation of phagocytosis when the cells were grown at 37° for 24–48 hr prior to HRG treatment. When macrophages were treated with 'human' HRG for longer than 8 hr, reduction of macrophage FcγR-mediated phagocytosis and binding also occurred, but with 10–20% less efficiency compared to homologous mouse HRG. Additionally, binding and/or possible phagocytosis of unopsonized E by macrophages via sialoadhesin receptors[21] remained at relatively low levels and unchanged for 1–24 hr in serum-free culture (Figs 2 and 3).

Differential regulation of FcγR subclasses

To further examine the function of FcγR subclasses, inflammatory macrophages were challenged with sheep erythrocytes coated with monoclonal IgG2a, IgG2b, or IgG1. Figure 4a shows that there was a dose-related reduction of FcγRII (Fcγ2b/γ1R)-mediated phagocytosis when the freshly harvested macrophages were pretreated with various quantities of HRG for 24 hr prior to interaction with EIgG2b or EIgGl. By contrast, under similar experimental conditions the FcγRI (Fcγ2aR)-dependent

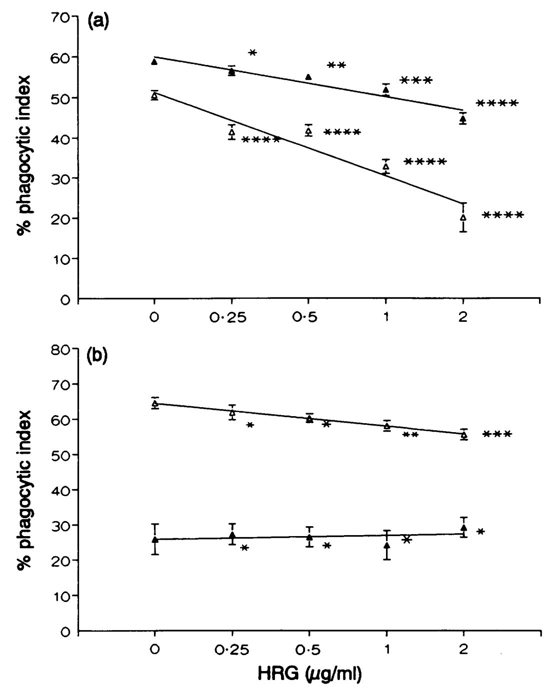

Figure 4. Effect of HRG on macrophage FcγR subclass-mediated phagocytosis. Freshly harvested inflammatory macrophages were treated with various quantities of HRG for 24 hr followed by assaying for phagocytosis of erythrocytes opsonized with (a) monoclonal IgG2b (1/10 dilution) (△) or IgG1 (1/8 dilution) (▲), or (b) IgG2a [1/25 (△) or 1/100 (▲) dilution] ($n=6$; mean ± standard deviation). Similar results were obtained using other antibody dilutions. *P*-values: experiments versus controls by Student's *t*-test: *$P>0{\cdot}05$; **$0{\cdot}01<P<0{\cdot}05$; ***$0{\cdot}001<P<0{\cdot}01$; ****$0{\cdot}01<P<0{\cdot}001$. Absolute values for the control groups in: (a) EIgG2b, 0·111 ± 0·003; EIgG1, 0·138 ± 0·004; E, 0·055 ± 0·002; and (b) EIgG2a—1/25, 0·218 ± 0·010; EIgG2a—1/100, 0·072 ± 0·004; E, 0·063 ± 0·001.

phagocytosis of EIgG2a was not effectively reduced by HRG (Fig. 4b). Similar results were obtained using sheep erythrocytes coated with other dilutions of IgG2a, IgG2b or IgG1 (data not shown).

Effect of HRG on unstimulated resident macrophages

Freshly harvested resident macrophages had low levels of binding and phagocytosis of EA cells (Fig. 5), compared to the inflammatory macrophages. However, following 18–24 hr in serum-free culture, there were increases in FcγR-dependent functions which corresponded to approximately both fourfold and threefold enhanced EA binding and phagocytosis, respectively (Fig. 5). HRG enhanced EA binding to macrophages following 1 hr pretreatment and subsequently reduced EA binding to macrophages during 18–24 hr treatment in a dose-dependent manner (Fig. 5). HRG also increased the phagocytic function of macrophages during 1–3 hr HRG pretreatment (Fig. 5b). Although there was an increased level of phagocytosis after control macrophages were in culture for 18–24 hr, HRG did not significantly reduce the FcγR-dependent phagocytosis ($P>0{\cdot}05$) (Fig. 5b).

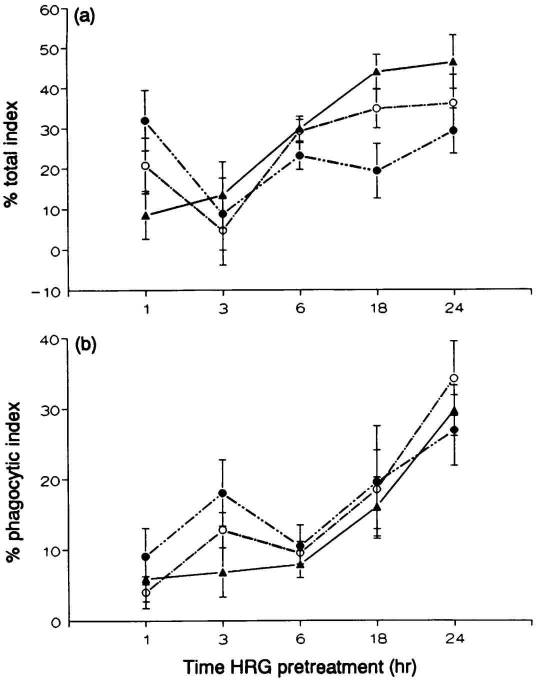

Figure 5. Effect of HRG on non-inflammatory resident macrophages. Peritoneal resident macrophages were treated with HRG [0 (▲); 100 (○); 200 (●) ng/ml] for 1–24 hr at 37°, followed by measuring the total index (a) of EA binding and phagocytosis, or the phagocytic index (b) of EA cells ($n = 4$; mean ± standard deviation). Absolute values for the control groups in: (a) EA, 0·097 ± 0·007, 0·130 ± 0·013, 0·174 ± 0·008, 0·171 ± 0·015, 0·230 ± 0·030, and E, 0·089 ± 0·009, 0·112 ± 0·010, 0·122 ± 0·009, 0·096 ± 0·003, 0·123 ± 0·012, corresponding to 1, 3, 6, 18 and 24 hr, respectively; and (b) EA, 0·064 ± 0·002, 0·066 ± 0·002, 0·070 ± 0·001, 0·075 ± 0·004, 0·091 ± 0·005, and E, 0·060 ± 0·001, 0·061 ± 0·001, 0·064 ± 0·001, 0·063 ± 0·003, 0·064 ± 0·002, corresponding to 1, 3, 6, 18 and 24 hr, respectively.

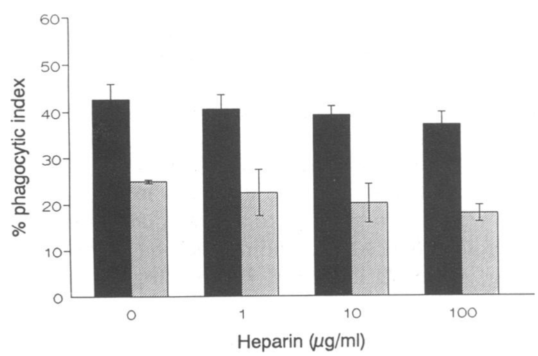

Figure. 6. Effect of heparin on HRG-mediated reduction of macrophage phagocytosis. Inflammatory macrophages were treated with (■) or without (■) HRG in the presence or absence of various concentrations of heparin for 24 hr at 37°, followed by testing for their phagocytic function ($n = 4$; mean ± standard deviation). Absolute values for the control groups: EA, 0·092 ± 0·006, E, 0·053 ± 0·001.

Effect of heparin and interferon-γ (IFN-γ)

Although heparin strongly binds HRG,[4,5] heparin failed to block the functional activity of HRG in reducing FcγR-mediated phagocytosis during 24 hr pretreatment of macrophages with heparin and/or HRG (Fig. 6). Similarly, IFN-γ, an essential stimulator for macrophage activation, did not block the function of HRG (data not shown).

DISCUSSION

In this study, it was demonstrated that HRG was capable of biphasic modulation of the expression of FcγR and its associated phagocytic function. HRG appeared to up-regulate rapidly the expression of FcγR during 1–2 hr pretreatment of inflammatory peritoneal macrophages. The increased FcγR expression was evidenced by the significantly enhanced EA binding to macrophages, coupled with enhanced phagocytosis. The opsonic activity of HRG in enhancing EA phagocytosis is unknown. However, the initial increased FcR function was probably not due to the opsonic activity of HRG, simply because HRG could not bind to erythrocytes,[13] and there were no significant differences in the results when the binding/phagocytosis assays were performed in the presence or absence of HRG. Moreover, HRG by itself did not act as a 'bridge' to enhance EA binding to macrophages, since HRG had no modulatory effect when the macrophages were treated with HRG for 1–2 hr at 4° followed by 1 hr co-incubation with EA at 37°. HRG down-regulated the FcγR expression on macrophages after 8–24 hr treatment, which resulted in significantly reduced EA binding and phagocytosis. To examine further which type of FcγR was affected by HRG, it was found that HRG down-regulated the expression of FcγRII of inflammatory macrophages and this corresponded to a dose-dependent decrease in the phagocytosis of erythrocytes opsonized with IgG2b or IgG1. The expression of FcγRI was less effectively affected by HRG.

Although these experiments were performed under serum-free conditions to prevent possible interactions between HRG and serum proteins or other factors,[2-8] the reduction of macrophage FcγR expression and phagocytic function was not due to loss of cell viability during the 24–48-hr culture period, as determined by ^{51}Cr-release assays. Indeed, there were increases in FcγR expression and phagocytic function for both resident and inflammatory macrophages during the 24-hr culture period without the presence of FBS. For example, the percentage of EA phagocytosis, E coated with 1/4000 diluted IgG, increased from 15% up to 48% when the freshly harvested inflammatory macrophages were cultured up to 24 hr under serum-free conditions.

HRG was found not to be cytotoxic to macrophages. A prolonged treatment of macrophages for more than 3 days with HRG resulted in an enhanced loss of spreading and adherence to plastic wells. Indeed, HRG has been determined to be a potent regulator for cellular protein synthesis and secretion in several cell types (N.-S. Chang, R. W. Leu, J. A. Rummage, J. K. Anderson and J. E. Mole, manuscript in preparation). The loss of adherence of HRG-treated macrophages is most likely associated with reduction of synthesis of cell-surface adhesive proteins. The sialic acid specific binding receptors, sialoadhesin, on macrophages contribute in part to the binding, but not phagocytosis, of E or EA to macrophages.[21] The level of sialoadhesin on macrophages during 24 hr in serum-free culture appeared to remain relatively steady. However, as an effective protein synthesis modulator, HRG may down-regulate the expression of sialoadhesin after 18–24 hr treatment and this contributes in part to a decreased binding of EA cells to either resident or inflammatory macrophages.

While in culture for 1–24 hr, non-inflammatory resident macrophages underwent differentiation upon plastic adherence, which was evidenced by their increased FcγR expression and phagocytic function. HRG reduced the expression of FcγR in a dose-dependent manner after 18–24 hr treatment of resident macrophages, thereby resulting in decreased EA binding. By contrast, the phagocytic function of resident macrophages was not affected by HRG. These results suggest that a specific FcγR subclass may be sensitive to HRG regulation of cell surface expression, but this FcγR subclass did not appear to contribute substantially to phagocytic function. Conceivably, the expression of FcγRII was down-regulated by HRG on resident macrophages. The FcγRII on resident macrophages may have a low phagocytic capacity, which requires a stimulatory signal in order to express fully its function. Indeed, it has been reported that FcγRI-mediated phagocytosis is generally far more efficient than that of FcγRII.[22,23] The low efficiency of FcγRII-mediated phagocytosis may be due to its intrinsic phospholipase A_2 activity, in which activation of phospholipase A_2 disrupts receptor and cytoskeletal protein interactions, that are essential for internalization.[24] Therefore, enhancement of FcγRII-mediated phagocytosis can be achieved by treating macrophages with inhibitors of phopholipase A_2 or cyclooxygenase.[25] Whether HRG activates or inhibits phospholipase A_2 will require further investigation.

Heparin has been shown to down-regulate the FcγRI-mediated phagocytosis of EIgG2a, but up-regulate FcγRII-mediated phagocytosis of EIgG2b.[26] Restriction of FcγRI phagocytic function by heparin is due to its regulation of FcγRI-associated casein kinase II activity and interaction with cytoskeletal proteins.[26] It was determined that heparin could not counteract the down-regulation of FcγR expression by HRG, even though heparin has a high affinity for binding to HRG. Differential regulation of FcγRI or FcγRII by heparin and HRG is unknown and remains to be investigated. However, the data suggest that HRG regulates macrophage function via a novel signal pathway different from that of heparin. Similarly, IFN-γ failed to block HRG function, suggesting that both molecules exerted different regulatory pathways in macrophages.

Molecular mechanisms by which HRG modulates macrophage FcγR expression and phagocytosis are not elucidated. Presumably, binding of HRG to specific receptors on macrophages would result in novel cellular responses. The post-receptor event is associated in part with regulation of cellular protein synthesis (N.-S. Chang, R. W. Leu, J. A. Rummage, J. K. Anderson and J. E. Mole, manuscript in preparation), with possible involvement of a nuclear transcription factor that controls basal transcription of genes (N. S. Chang, unpublished observations). Although the physiological function of HRG in serum is not well understood, it is believed to act as a negative acute phase reactant.[27] This observation implies that reduced HRG concentration in blood during an acute phase response may be critical for maximizing cell functions, as well as the functions of serum components. Additionally, a balanced concentration of HRG in plasma may be of utmost significance. For example, an elevation of plasma HRG may contribute to thrombophilia as indicated by several clinical findings.[28] It was reported recently that the balanced concentration of HRG in plasma appears to be necessary for optimizing complement function.[8] Taken together, the observations in this study on HRG modulation of macrophage function, along with other findings on its regulation of T-cell functions,[10–13] further establish the role of HRG in modulation of immune cell functions.

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